

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Determine Equilibriums of 1st-Order ODEs

1. Exponential Growth

In this problem, we study the rate of change equation designed to inform us about the future populations of a species. We have the following assumptions: The species has been in its habitat for some time, the species has access to unlimited resources (e.g. food, space, water), and the species reproduces continuously.

The population change of the species with the given assumptions can be modeled using an ODE given as

$$P' = \frac{4}{5}P$$

where P is the population of the species at time t and $\frac{4}{5}$ is known as the growth constant.

- a. The general solution of the given ODE is

$$P(t) = Ce^{\frac{4}{5}t}$$

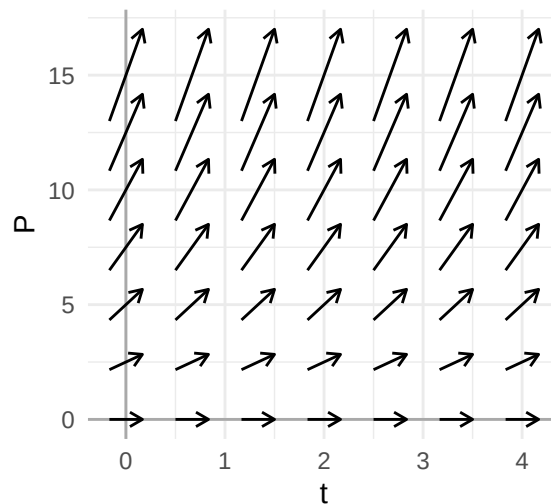
where C is an arbitrary constant. Compute the specific solutions for each initial condition.

i. $P(0) = 1$

ii. $P(0) = 5$

- c. On the initial condition of $P(0) = 0$, the specific solution is $P(t) = 0$. Show that this is also an equilibrium solution.

- b. Use the slope vectors on the slope field shown below to sketch the specific solutions from Part (a) and identify any equilibrium solutions.



- d. How does the general solution differ from a specific solution in the context of the slope field? Explain your reasoning.

2. Logistic Growth

In the previous problem, we saw that the given ODE can be used to model a situation where there is one species, continuous reproduction, and unlimited resources. In most situations, however, the resources are not unlimited, so to improve the model one has to modify the ODE to account for the fact that resources are limited.

The population change of the species with the updated assumptions can be modeled using a new ODE given as

$$P' = \frac{4}{5} \left(P - \frac{P^2}{10} \right)$$

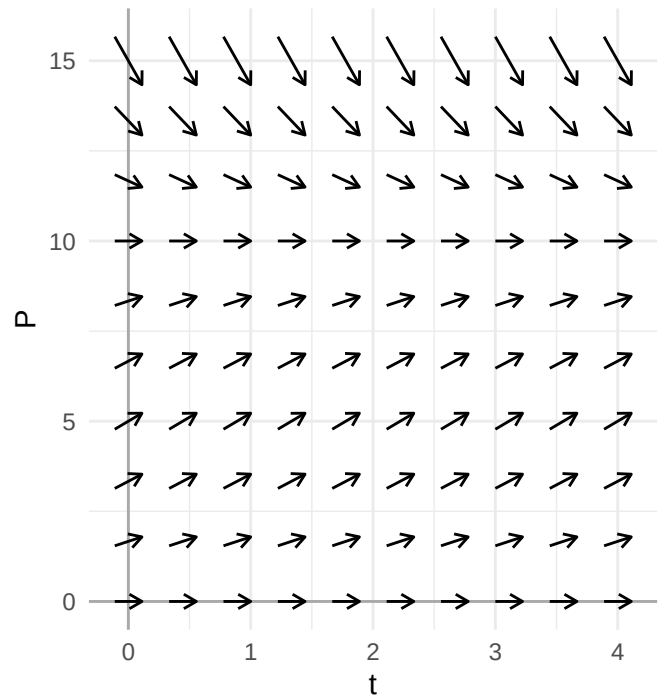
where the P is the population of the species at time t , $\frac{4}{5}$ is the growth constant, and 10 is the carrying capacity.

- a. Compute all of the equilibrium solutions, $P(t) = P^*$.

- c. Show that the solutions determined in Part (a) satisfies the given ODE.

- d. Show that the solutions are increasing if the initial condition is $0 < P(0) < 10$, and decreasing if the initial condition is $P(0) > 10$.

- b. Use the slope vectors on the slope field shown below to sketch at least three specific solutions and identify any equilibrium solutions.



- e. How does the solutions behave when it starts above and below the carrying capacity? What does this mean in context of population growth/decay? Explain your reasoning.